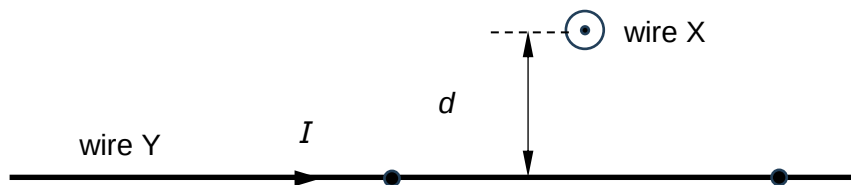


- 23** Two long straight wire X and Y are placed perpendicular to each other at a distance d apart. A current flows out of the page in wire X. The same current flows from left to right in wire Y.

What are the directions of the forces acting on wire Y at the point P and Q due to the magnetic field produced by wire X?



	force at P	force at Q
A	out of page	into page
B	into page	out of page
C	towards X	away from X
D	towards X	towards X

Ans: B

Using right-hand grip rule for the wire X, the B-field due to wire X is anti-clockwise on the plane of paper cutting through point P and Q, giving a vertical component downwards at point P, and vertical component pointing upwards at point Q.

Using Fleming's Left Hand rule at point P and Q, the direction of magnetic force can be obtained at points P and Q.

24 As a rectangular coil of dimensions l by l enters a magnetic field of flux density B at right